

Semiparametric shrinkage Bayes estimators of high-dimensional data

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Abstract

In this article, due to the large number of data, the model is considered semi-parametric in order to keep the advantages of parametric and non-parametric models together. With the advancement of science, knowledge, and technology, new and comprehensive methods have been invented to measure, collect and record data, that in turn has led to the emergence and development of high-dimensional data, in which the number of covariates is much larger than the number of samples. In addition, the data have changed in terms of structure, and these changes can no longer be analyzed with traditional and classic methods, resulting in the complexity of their interpretation, which demands the use of new methods. We use from the semi-parametric model the use of penalty functions for high-dimensional data. Covariates from different and multiple sources are considered, and Empirical Bayes and Hierarchical Bayes methods are combined to estimate parameters and hyperparameters with an additional level of flexible shrinkage. This provides a unique framework because it can use any type of shrinkage. In our proposed method, Empirical Bayes and Hierarchical Bayes methods in the semi-parametric model was create a different approach because they use high-dimensional and unstructured data. For simulated data and an applications example of cervical cancer data, we demonstrate improvements in terms of performance compared to previous methods.

Keywords: Co-data, semi-parametric models, penalty functions, high-dimensional data, empirical Bayes, hierarchical Bayes.

1 Introduction

Choosing the variable and estimating the coefficients in the regression model is the most basic part in modeling. So far, various criteria and methods have been presented for variable selection, the most famous of which are Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Contrary to

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the widespread use of these criteria in the field of variable selection, their use causes instability in variable selection. This is because if we use these criteria, we can only make two decisions about each regression coefficient. One way to overcome this problem is to use penalty functions for variable selection. On the other hand, the estimation methods of the least square or rigid regression method, as well as the method of selecting the leading variable, do not show significant performance when faced with data that have different characteristics. Instability, low prediction accuracy, and incorrect selection of variables are examples of model damages when using these methods. In addition, these damages are aggravated when the correlation between predictor variables is high.

shrinkage estimations have been regarded as a way to reduce these damages when the correlation between predictor variables is high. In this regard, the efficiency of shrinkage estimators for the error squared loss function increases. In these methods, regression coefficients are estimated by applying restrictions on the variation range. Such a limitation reduces the variance of the estimator, but it will create bias for the estimator; consequently, it can be hoped that the mean square error has decreased. In general, to solve such problems, the bias can be slightly increased through shrinking some of the coefficients of the estimators or setting them to zero, but the variance can accordingly be reduced. The main solution for this is variable selection with penalty function. Tibshirani (1996) introduced the penalty function called least absolute shrinkage and selection operator (LASSO) and Fan and Li (2001) introduced the smoothly clipped absolute deviation (SCAD) penalty function. In the presence of these penalty functions, variable selection and parameter estimation are performed simultaneously. They stated the characteristics of a good penalty function are (1) bias, (2) tightness and (3) continuity.

In recent years, structured data has undergone fundamental changes in terms of diversity and dimensions, and these changes have been included in different studies. This type of data, in terms of variety and dimension, is not compatible with current conventional methods used in various articles. It is necessary to propose a new structure for these studies to create valid results for them. Also, nowadays, structure data such as unstructured, structured, pseudo-structured, and semi-structured data are more diverse, which should be used in research, and the analysis of these data can no longer be efficient and useful using the previous conventional methods. As a result, in recent years, different studies have been conducted on high-dimensional data or on different types of data that have grown significantly under the title of big data. The method recently introduced by Vannee et al. (2021) is the ecpc * method, which works on high-dimensional data and parametric linear models. Typically, in this method, high-dimensional, unstructured data are also modeled using prior information, statistical prediction, and variable selection by choosing a specific penalty (or prior) that penalizes (or shrinks) the model parameters.

*empirical Bayes Co-data learnt Prediction and Covariate selection

Nowadays, due to the large volume of medical, engineering and environmental data, the need for flexible statistical models such as semi-parametric regression models has increased. This is because through these models, it is possible to analyze complex data that are being developed. Semiparametric regression models combine the virtues of parametric and nonparametric modeling. Semi-parametric regression includes two components: checking the model for non-parametric components and checking the model for the parametric components. Therefore, these investigations have more challenges than parametric models such as linear models and generalized linear models. Bayesian, empirical Bayes and hierarchical Bayes approaches are used to solve most statistical problems. These methods have a useful feature that, if used correctly, can be very efficient for predicting and selecting variables using penalties. The methods used in this field can be extended for the multivariate case, but it may not be suitable and useful for high-dimensional data in practice.

Lasso, which is one of the most popular penalties performed by [Tibshirani \(1996\)](#), ridge regression, which was carefully considered by [Hoerl and Kennard \(1970\)](#), elastic Net, which was first reviewed by [Zou and Hastie \(2005\)](#) and developed by [Hastie and Tibshirani \(2020\)](#). [Vandewie et al. \(2016\)](#) developed a method for adaptive group-regularized ridge regression, which makes structural use of co-data. They divide "groups" into partitions of variables based on co-data and use an empirical Bayes estimation approach, called GRidge. [Velten and Huber \(2019\)](#) worked on high-dimensional data using Adaptive Penalization in High-Dimensional Regression and Covariate Classification using Bayesian. The gren method by [Münch et al. \(2019\)](#) was performed with group-adaptive elastic net priors. In the fwelnet package introduced by [Tay et al. \(2020\)](#), the regression coefficients are jointly estimated with co-data variable weights, and variable-specific elastic net penalties are modeled by a normalized exponential function and an exponential of the co-data. [Vannee et al. \(2022b\)](#) provide s fast approximate marginal likelihood estimates s for group-adaptive elastic net penalties, but grouped co-data only. [Ryan and Farshidi \(2021\)](#) introduced a class of group sparse estimators that combine group subset selection with group lasso or ridge shrinkage and develop an optimization framework for non-convex regularization surface fitting. As an application for group selection, they chose sparse semi-parametric modeling. They showed the efficiency of new estimators in modeling for economic problems.

In Bayesian theory and empirical Bayesian approach to high-dimensional data analysis, when the number of covariates is very large, we will have at least one covariate with a nonparametric coefficient. In addition, due to the large volume of data in various fields, the need for flexible statistical models such as semi-parametric regression models has increased. As a result, We focus on semiparametric regression models in unstructured and high-dimensional data. To estimate the parameters in the parametric part of the semi-parametric regression model, we use the empirical Bayes approach with the penalized likelihood function,

and in the non-parametric part, we use the hierarchical Bayes method in the presence of the estimated parameters of the parametric part. To estimate multigroup adaptive penalties, we use penalized ridge generalized linear models with different and multiple types of shared data/co-data.

This paper is organized as follows. Section 2 presents the symbols and models introduced in this paper. Section 3 presents the estimation of the parameters in the semi-parametric model. Some asymptotic theorems in semiparametric model and ecpc method are introduced in Section 4. Section 5 presents a simulation study that shows how the additional shrinkage layer in the semiparametric model shrinks the group weights. Then Section 6 examines the proposed model in a real example for cervical cancer genomic data and compares it with previous works. Finally, Section 7 concludes and discusses the method.

2 Method

2.1 Notation

We denote the response vector with $\mathbf{Y} \in \mathbb{R}^n$ and the data matrix with $X \in \mathbb{R}^{n \times p}$, $Z \in \mathbb{R}^{n \times q}$, and $p \gg n$. We use D different co-data sources to define groups of covariates. The collection of all groups of the d th co-data source is showed in a group set. D group set with the symbol of $\mathfrak{g}^{(d)}$ is shown in such a way that $G^{(d)} := |\mathfrak{g}^{(d)}|$, and we also denote the g group of the d th group set with the symbol $\mathfrak{g}_g^{(d)}$, $G_g^{(d)} := |\mathfrak{g}_g^{(d)}|$. D the design matrix is defined to represent a group member of the co-data group set d .

Definition 2.1. *It is assumed that k is a group member of $|I_k^{(d)}|$ groups in group set d in such a way that $I_k^{(d)}$ represents a set of functions indicating the groups in group set d of member k . We define the design matrix or co-data matrix with the symbol $T^{(d)} \in \mathbb{R}^{p \times G^{(d)}}$ the set of groups corresponding to d , with the elements $[T^{(d)}]_{kg}$ for the k th covariate and the g th group as follows*

$$[T^{(d)}]_{kg} = \begin{cases} \frac{1}{|\mathfrak{J}_k^{(d)}|} & \text{if } g \in \mathfrak{J}_k^{(d)} \\ 0 & \text{if not} \end{cases}, d = 1, \dots, D, \quad k = 1, \dots, p, \quad g = 1, \dots, G$$

2.1.1 Semi-parametric model for co-data and high-dimensional data

If Y is the response variable and Z , X are its related variables, the semi-parametric model based on the non-parametric model of [Li and Liang \(2008\)](#) with parameters β and α is considered as follows:

$$E(g\{X, Z\}) = X\beta + \alpha(Z) \tag{1}$$

Where $\beta \in \mathbb{R}^p$ is a vector of regression coefficients and $\alpha(\cdot)$ is a vector consisting of smoothing regression coefficient functions. Model (1) is a semi-parametric model, $X\beta$ is the parametric component and $\alpha(Z)$

is the non-parametric part. Let us denote the corresponding log-likelihood by $\ell(\beta, \alpha)$ and the penalized log-likelihood by

$$\ell(\beta, \alpha) = l(\beta, \alpha) - \sum_{k=1}^p \frac{1}{\tau_{(k, local)}^2} \beta_k^2$$

It is assumed that β has a prior normal distribution with prior variance $\tau_{k, local}$. If z_f is the i th column of matrix Z , $\alpha(Z) = \{\alpha(z_1), \alpha(z_2), \dots, \alpha(z_q)\}$ has a normal distribution with zero mean and variance $cov(\alpha(z), \alpha(z')) = \tau K(z, z')$ in such a way that τ is an unknown positive parameter and $K(., .)$ is a kernel function and an $n \times n$ matrix with the ff' th member $K(z_f, z_{f'})$, $f, f' = 1, 2, \dots, q$. Therefore, the model is as follows:

$$\begin{aligned} Y_i | X_i, Z_i, \beta, \alpha &\sim N(X_i \beta + \alpha(Z_i), \sigma^2 I) \quad i = 1, 2, \dots, n \\ \beta_k | \tau_{(k, local)}^2 &\sim N(0, \tau_{(k, local)}^2) \quad k = 1, 2, \dots, p \\ \tau_{k, local}^2 &= \tau_{global}^2 \sum_{d=1}^D w^{(d)} T_k^{(d)} \gamma^{(d)} \\ \alpha_f(Z_i) | \tau &\sim N(0, \tau K(z, z')) \quad f = 1, 2, \dots, q \\ \tau &\sim IG(a_\tau, 1/b_\tau) \\ \sigma^2 &\sim IG(a_{\sigma^2}, b_{\sigma^2}) \end{aligned} \tag{2}$$

Where X_i and T_k are respectively the i th and k th rows of the matrix X and T and $a_\tau, b_\tau, a_{\sigma^2}$ and b_{σ^2} are fixed values for the hyperparameters σ^2 and τ . $\tau_{k, local}^2$ represents local prior variance on co-data of $T^{(d)}$, $d = 1, 2, \dots, D$, $\gamma^{(d)} \in \mathbb{R}_+^{G^{(d)}}$ represents the group weight vector, and $w^{(d)} \in \mathbb{R}_+$ represents the group set weight. In this article, for the non-parametric part, the following kernels are used for K :

- (1) The polynomial approach: The polynomial kernel function of r th, as $K(z, z') = (zz' + \rho)^r$, $r = 1, 2$, where $\rho = 0$ if $r = 1$ and $\rho = 1$. If $r = 2$,
- (2) Gaussian kernel function: $K(z, z') = \exp(-\frac{\|z - z'\|^2}{\rho})$, where $\|z - z'\|^2 = \sum_{k=1}^p (z_k - z'_k)^2$. In this article, ρ is assumed to be a known scale parameter.

3 Estimate

The unknown parameters are the semiparametric regression coefficients, β and α , and hyperparameters (i.e., $\Psi = \{\tau, \sigma^2, \gamma^{(1)}, \dots, \gamma^{(d)}, w^{(d)}, \dots, w^{(d)}\}$). As $\tau_{k, local}$ is a linear combination of other unknown parameters, we do not consider this parameter. In this article, the empirical Bayes approach is used to estimate the unknown parameters $\{\beta, w^{(1)}, \dots, w^{(d)}, \gamma^{(1)}, \dots, \gamma^{(d)}\}$, and so is the hierarchical Bayes approach to estimate the unknown parameters $\Psi_1 = \{\alpha, \sigma^2, \tau\}$. Finally, the estimated hyperparameters are placed in the following

penalized likelihood function to be estimated by maximizing β at each $d = 1, \dots, D$ group.

$$\operatorname{argmax}_{\beta, \alpha} \left\{ \log \pi_{\hat{\sigma}, \hat{\tau}, \hat{\gamma}^{(d)}, \hat{w}^{(d)}}(\mathbf{Y} | \mathbf{X}, \mathbf{Z}, \alpha, \beta) - \sum_{k=1}^p \frac{1}{\hat{\tau}_{k,local}^2} \beta_k^2 \right\} \quad (3)$$

where $\hat{\sigma}$, $\hat{\tau}_{k,local}^2$, $\hat{\tau}$, $\hat{\gamma}^{(d)}$ and $\hat{w}^{(d)}$ respectively are the estimates of parameters of σ , $\tau_{k,local}^2$, τ , $\gamma^{(d)}$ and $w^{(d)}$. Of course, it should be noted that the parameter ρ is assumed to be constant in the case where the kernel function is Gaussian. Therefore, the Newton-Raphson method is used to maximize the relationship (3). The relation (3) is actually an ordinary ridge regression with a weight penalty that can be easily solved with existing software such as glmnet R package. As model (2) has two parametric and non-parametric parts, we estimate the parameters in the parametric and non-parametric parts in the semi-parametric models separately with the empirical Bayes approach and the hierarchical Bayes approach as follows:

Step(1) : Empirical Bayes estimation of the parametric component β_k

Step(2) : Hierarchical Bayes estimation of the non-parametric component $\alpha_f(Z_i)$ considering the known parametric component

Step(3) : Empirical Bayes estimation to estimate parameters, $\gamma^{(d)}$, $\tau_{k,local}^2$, $w^{(d)}$ and β

3.1 Empirical Bayes estimation for the parametric component β_k

First, we solve a part of the non-parametric expression in the model. Based on $h > 0$, which is called the bandwidth, the Nadaraya Watson kernel estimate for the b th element of $\alpha^{(0)}(Z) = (\alpha^{(0)}(z_1), \dots, \alpha^{(0)}(z_f))$, $f = 1, 2, \dots, q$, is defined as follows

$$\alpha^{(0)}(z_f) = \sum_{i=1}^n l_i(z_f) Y_i \quad (4)$$

where the weights of $l_i(z_f)$ are as follows

$$l_i(z_f) = \frac{K\left(\frac{z_f - z_i}{h}\right)}{\sum_{i=1}^n K\left(\frac{z_f - z_i}{h}\right)}$$

Our aim is to obtain the estimate of β with respect to the initial value of $\alpha^{(0)}$. The function of the logarithm of the likelihood of the penalty at this stage is as follows by replacing in the relation (2)

$$L(\alpha^{(0)}, \beta) = \sum_{i=1}^n (Y_i - X_i \beta - \alpha^{(0)}(Z))^2 - \sum_{k=1}^p \frac{1}{\tau_{k,local}^2} \beta_k^2 \quad (5)$$

Normally, Newton-Raphson algorithm is used to maximize (5). We assume that the initial value of $\beta^{(0)}$ is as follows

$$\beta^{(0)} = X' X^{-1} X' Y$$

To maximize the relation (5), we use Taylor's first order approximation. If $\hat{\beta}$ is the smoothed first approximation of β , then the first order approximation of $\hat{\beta}$ is as follows

$$\begin{aligned}
\hat{\beta} &\approx \beta - \left[\frac{\partial^2 \ell(\alpha^{(0)}, \beta)}{\partial \beta \partial \beta'} \right]^{-1} \frac{\partial \ell(\alpha^{(0)}, \beta)}{\partial \beta} \\
&= \beta - \left[-X' \sigma^{-2} X - \frac{2}{\tau_{k,local}^2} \right]^{-1} \left[X'(Y - X\beta - \alpha^{(0)}(Z)) - \frac{2}{\tau_{k,local}^2} \beta \right] \\
&= \left[X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right]^{-1} \left[X'(Y - X\beta - \alpha^{(0)}(Z)) - \frac{2}{\tau_{k,local}^2} \beta + (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2}) \beta \right] \\
&= \left[X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right]^{-1} \left[X'(Y - X\beta - \alpha^{(0)}(Z)) + X' \sigma^{-2} X \beta \right]
\end{aligned}$$

Based on the characteristics of penalties, the β estimator is biased, that is, $E_{Y|\beta}(\hat{\beta}_k) \neq \beta_k$. Roughly, $E_{Y|\beta}(\hat{\beta}_k)$ and $V_{Y|\beta}(\hat{\beta}_k)$ are calculated below. First order approximation $E_{Y|\beta}(\hat{\beta}_k)$

$$\begin{aligned}
E_{Y|\beta}(\hat{\beta}_k) &= E_{Y|\beta} \left(\left[X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right]^{-1} [X'(Y - X\beta_k - \alpha^{(0)}(Z)) + X' \sigma^{-2} X \beta] \right) \\
&= \left[X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right]^{-1} [X'(E_{Y|\beta}(Y) - X\beta_k - \alpha^{(0)}(Z)) + X' \sigma^{-2} X \beta] \\
&= \left[X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right]^{-1} [0 + X' \sigma^{-2} X \beta_k] \\
&= \left[X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right]^{-1} + X' \sigma^{-2} X \beta_k = \sum_{l=1}^p c_{kl} \beta_k
\end{aligned}$$

Also for $V_{Y|\beta}(\hat{\beta}_k)$

$$\begin{aligned}
\hat{\Sigma} = V_{Y|\beta}(\hat{\beta}_k) &= V_{Y|\beta} \left(\left[X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right]^{-1} [X'(Y - X\beta_k - \alpha^{(0)}(Z)) + X' \sigma^{-2} X \beta] \right) \\
&= \left(X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right)^{-1} X' V_{Y|\beta}(Y) X \left(X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right)^{-1} \\
&= \left(X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right)^{-1} X' \sigma^2 I X \left(X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right)^{-1} \\
&\approx \left(X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right)^{-1} X' \hat{\sigma}^2 I X \left(X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2} \right)^{-1}
\end{aligned}$$

Calculation of $E_{Y|\beta}(\hat{\beta}_k)$ and $\hat{\Sigma}$ requires the inverse of the $p \times p$, matrix, $M_{\tau_{k,local}^2} = (X'X + \frac{2}{\tau_{k,local}^2})^{-1}$. Therefor The singular decomposition of $X' = UDV'$ reduces the calculation of the inverse of $[M]_{\tau_{k,local}^2}$ to an $n \times n$ inverse and matrix multiplication. $[M]_k$ represents the kth row (component) of each matrix (vector) M.

According to the definition ??, considering that the co-data is collected from D different sources and each one is a partition of the variables into $g^{(d)}$ groups, so the penalty term in (5) is generalized as follows

$$\sum_{i=1}^n \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ \frac{1}{2\sigma^2} (Y_i - X_i\beta - \alpha^{(0)}(Z))^2 \right\} \right] - \sum_{g=1}^{G^{(d)}} \sum_{k=1}^p \frac{1}{\tau_{kg,local}^2} \beta_k^2 \quad (6)$$

Then, prior mean and prior variance can be estimated by solving moment equations if the conditional moments $E_{Y|\beta}(\hat{\beta}_k)$ and $E_{Y|\beta}(\hat{\beta}_k^2)$ are analytically tractable as functions of β_k .

We have $E_{Y,\beta}(\hat{\beta}_k^2) = V_{Y,\beta}(\hat{\beta}_k)$ and $E_{\beta_i,\beta_j}(\beta_i, \beta_j) = 0$. . As a result,

$$\begin{aligned}
\forall g = 1, \dots, G^{(d)} : E_{Y,\beta}(\hat{\beta}_k^2) &= E_\beta[V_{Y|\beta}(\hat{\beta}_k) + (E_{Y|\beta}(\hat{\beta}_k))^2] \\
&= E_\beta[V_{Y|\beta}(\hat{\beta}_k)] + V_\beta[E_{Y|\beta}(\hat{\beta}_k)] + E_\beta(E_{Y|\beta}(\hat{\beta}_k))^2 \\
&= E_\beta[(X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} X' \hat{\sigma}^2 I X (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1}] \\
&\quad + V_\beta[(X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} + X' \sigma^{-2} X \beta_k] \\
&\quad + E_\beta([X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} + X' \sigma^{-2} X \beta_k)^2 \\
&= (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} X' \hat{\sigma}^2 I X (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} \\
&\quad + (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} X' \sigma^{-2} X V_\beta[\beta_k] \\
&\quad \times X' \sigma^{-2} X (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} \\
&\quad + (X' \sigma^{-2} X \beta_k)^2 \\
&= (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} X' \hat{\sigma}^2 I X (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} \\
&\quad + (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} X' \sigma^{-2} X \tau_{k,local}^2 \\
&\quad \times X' \sigma^{-2} X (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} \\
&\quad + (X' \sigma^{-2} X \beta_k)^2
\end{aligned} \tag{7}$$

We can rewrite the relation (7) using the matrix C and ν_k as a function of the Pebshin variance $\tau_{k,local}^2$

$$\begin{aligned}
C &:= [X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2}]^{-1} X' \sigma^{-2} X \\
\hat{\nu}_k &:= (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} X' \hat{\sigma}^2 I X (X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2})^{-1} \\
E_{Y,\beta}(\hat{\beta}_k^2) &= \hat{\nu}_k + C^2 \tau_{k,local}^2 + (X' \sigma^{-2} X \beta_k)^2
\end{aligned} \tag{8}$$

3.2 Hierarchical Bayes estimation for non-parametric $\alpha_f(Z_i)$

We assume that the non-parametric function $\alpha(Z)$ lies inside a function space H_K which is created by a positive definite kernel function $K(\cdot, \cdot)$. From Mercer's theorem (Cristianini and Shawe-Taylor (2006)), under certain order conditions, a kernel function $K(\cdot, \cdot)$ implicitly defines a unique function space covered by a certain set of orthogonal basis functions $\varphi_f(z)_{f=1}^q$. In other words, each $\alpha(z) \in H_K$ can be represented using a set of bases such as $\alpha(z) = \sum_{f=1}^q \omega_f \varphi_f(z) = \varphi(z)^T \omega$, where ω is the vector of coefficients.

The function space considered in this article is as follows: (1) rth polynomial kernel: $K(z_1, z_2) = (z_1^T z_2 + \rho)^r$

where ρ and r are the tuning parameters. The r th polynomial kernel creates the function space spanned by all possible r th monomials, which are components of Z . For example, if $d = 1$, the first polynomial kernel generates a linear function space with basis functions $\varphi_f(z) = z_f, \dots, z_q$. If $d = 2$, the second polynomial kernel corresponds to the second function space with basis functions $\varphi_f(z) = z_f, z_f z'_f (f = 1, \dots, q)$. (2) the Gaussian kernel: it creates a function space covered by radial basis functions. See [Buhman \(2003\)](#) for their mathematical and desirable properties.

To estimate the parameters α , τ and σ^2 we use the following hierarchical Bayes, and under the Gaussian function and the polynomial kernel function and the fact that $\hat{\beta}$ is known, we seek to estimate the parameter Ψ_1

$$Y_i | X_i, Z_i, \hat{\beta}, \alpha \sim N(X_i \hat{\beta} + \alpha(Z_i), \sigma^2 I) \quad i = 1, 2, \dots, n$$

Marginal distribution of Y

$$\begin{aligned} [Y | X, Z, \alpha(Z), \Psi] [\alpha(Z) | \Psi] &\sim \frac{1}{(2\pi)^n (\sigma^2)^{n/2} \tau^{n/2} |K|} \\ &\times \exp\left\{-\frac{(Y - (X\hat{\beta} + \alpha(Z)))' (Y - (X\hat{\beta} + \alpha(Z)))}{2\sigma^2}\right\} \\ &\times \exp\left\{-\frac{\alpha(Z)' K^{-1} \alpha(Z)}{2\tau}\right\} \end{aligned}$$

And complete conditional distributions are as follows

$$\begin{aligned} [\sigma^2 | Y, X, Z, \alpha(Z), \rho, \tau, \hat{\beta}] &\sim IG\left(\frac{n}{2} + a_{\sigma^2}, \frac{(y - (X\hat{\beta} + \alpha(Z)))' (y - (X\hat{\beta} + \alpha(Z)))}{2} + b_{\sigma^2}\right) \\ [\alpha(Z) | Y, X, Z, \sigma^2, \rho, \tau, \hat{\beta}] &\sim N[\sigma^{-2} \{(\tau K)^{-1} + \sigma^{-2} I\}^{-1} (y - X\hat{\beta}), \{(\tau K)^{-1} + \sigma^{-2} I\}^{-1}] \quad (9) \\ [\rho | y, X, Z, \alpha(Z), \sigma^2, \tau, \hat{\beta}] &\propto \left(\frac{1}{(2\pi)^{n/2} |\tau K(\rho)|^{1/2}} \exp\left\{-\frac{\alpha(Z)' K(\rho)^{-1} \alpha(Z)}{2\tau}\right\}\right) \end{aligned}$$

$$[\tau | y, X, Z, \alpha(Z), \rho, \sigma^2, \hat{\beta}] \sim IG\left(\frac{n}{2} + a_{\tau}, \left\{\frac{\alpha(Z)' K(\rho)^{-1} \alpha(Z)}{2} + b_{\tau}\right\}^{-1}\right)$$

Since complete conditional distributions for all parameters have a closed form, MH algorithm and Gibbs sampling are used in the joint posterior distribution. We need to be more careful about K because if K is a non-singular matrix, we can directly sample their complete conditional distributions. However, if K is singular, we sample $[\alpha(Z) | rest]$ from the MH algorithm with the proposed distribution

$$N[\sigma^{-2} \{(\tau \{\delta I + K(\rho)\})^{-1} + \sigma^{-2} I\}^{-1} (Y - X\hat{\beta}), \{(\tau \{\delta I + K(\rho)\})^{-1} + \sigma^{-2} I\}^{-1}]$$

, with a very small value of δ with the uniform distribution in a small range like $\delta \in [0.0001, 0.01]$, and we sample $[\tau | rest]$ with the proposed distribution

$$IG\left(\frac{n}{2} + a_{\tau}, \frac{\alpha(Z)' \{\delta I + K(\rho)\}^{-1} \alpha(Z)}{2} + b_{\tau}\right)$$

with the same value of δ .

3.3 Empirical Bayes estimation to estimate parameters $\gamma^{(d)}$, $w^{(d)}$, $\tau_{k,local}^2$ and β

In this section, by replacing the parameters estimated in the previous step in relation (5), we estimate the parametric component and other remaining parameters with the empirical Bayes method of moments (MoM). The steps can be summarized as follows,

- (1) τ_{global}^2 : For semiparametric regression with a known $\hat{\alpha}(Z)$, we maximize the marginal likelihood directly because of its being analytic and set all local variances to 1. This parameter is first estimated by ignoring groups and group sets. Therefore, every β_k (1) has a normal distribution with the same variances.

$$\tau_{global} = \sqrt{\frac{\pi}{2}} E_{(\beta_k | \tau_{global}^2, \hat{\alpha}(Z))}(|\beta_k|) \approx \sqrt{\frac{\pi}{2}} \frac{1}{p} \sum_{k=1}^p |\beta_k|$$

- (2) Group weights for each group set, $\gamma^{(d)}$, $d = 1, \dots, D$, with a known $\hat{\tau}_{global}^2$: It uses the penalized moment-based estimates based on the initial estimate and the ordinary ridge $\hat{\beta}$. For this purpose, the adjustment of moment-based estimation equations in the structure of groups is used when the number of groups approaches or exceeds the number of samples. For different types of co-data, different penalty functions with hyperpenalty λ can be used, which need to be estimated at this stage. This is done in a data-driven way using group segmentation.
- (3) Group set weight, $W = (w^{(1)}, \dots, w^{(d)})$, $d = 1, \dots, D$, with a known $\hat{\gamma}^{(1)}, \dots, \hat{\gamma}^{(d)}$ and $\hat{\tau}_{global}^2$: This section is also used with moment-based estimates for the group set weight.

3.3.1 Group weights for each group set, $\gamma^{(d)}$, $d = 1, \dots, D$

We will continue to estimate the group weights using the empirical Bayes method of moments (MoM). According to equation (2) τ_{global}^2 is a linear combination of group weights $\gamma^{(d)}$

$$\begin{aligned} \forall g = 1, \dots, G^{(d)} \quad &: E_{Y, \beta}(\hat{\beta}_k^2) = E_{\beta}(E_{Y | \gamma^{(d)}, \hat{\tau}_{global}^2, \hat{\alpha}(Z)}(\hat{\beta}_k^2)) \\ &= \frac{1}{|p|} \sum_k^p E_{\beta | \gamma^{(d)}, \hat{\tau}_{global}^2, \hat{\alpha}(Z)} E_{Y | \beta}[\hat{\beta}_k^2 | \beta][\hat{\beta}_k^2[Y, \hat{\alpha}] | \beta] \\ &= \frac{1}{|p|} \sum_k^p h(\gamma^{(d)}) \end{aligned} \tag{10}$$

The function h in the equation (10) is a linear combination of $\gamma^{(d)}$. Equation (8) can be rewritten as linear system. that is, the solution of the moment estimation equations is reduced to the solution of a linear system with the number $G^{(d)}$ of the following equation:

$$A^{(d)} \gamma^{(d)} = b^{(d)}$$

$$\begin{aligned}
A^{(d)} &:= C^2 \hat{\tau}_{gloabl}^2 \sum_{d=1}^D w^{(d)} T_k^{(d)} \\
b^{(d)} &:= E_{Y,\beta}(\hat{\beta}_k^2) - \hat{\nu}_k - [X' \sigma^{-2} X]^2
\end{aligned} \tag{11}$$

Where $A^{(d)} \in \mathbb{R}^{G \times G}$ and $b^{(d)} \in \mathbb{R}^{G^{(d)}}$ depend on the X data and the initial estimate $\hat{\beta}_k^2(Y, \hat{\alpha}(Z), \hat{\tau}_{gloabl}^2)$.

For a limited number of non-overlapping groups of equal size, it suffices to solve the direct solution of the linear system leads to 0 by truncating the negative of the group weights estimates, which may be due to approximations or numerical errors. Given that there are many groups that are not equal in size, or have a hierarchical structure, we require penalizing the system. For example, in the simulation section, we show how ridge shrinkage can be used to avoid overfitting in many groups and how hierarchical lasso hypershrinkage can be used to select groups in a group set in hierarchical groups. Therefore, to solve the equation (11) it is suggested to minimize the least squares of penalty of the following relation for $\hat{\gamma}^{(d)}$.

$$\hat{\gamma}^{(d)} = (\tilde{\gamma}^{(d)})_+, \tilde{\gamma}^{(d)} = \operatorname{argmin}_{\gamma^{(d)}} \|A^{(d)} \gamma^{(d)} - b^{(d)}\|_2^2 + f_{pen}^{(d)}(\gamma^{(d)}; \hat{\lambda}^{(d)}) \tag{12}$$

Where $(\cdot)_+ = \max(0, \cdot)$, which is called the cut of the element of a vector at 0, and $\hat{\lambda}^{(d)}$, the hyperparameter estimate $\lambda^{(d)}$, which is explained below. Note that the solution of equation (??) corresponds to the solution of penalized linear regression with the penalty function $f_{pen}^{(d)}$, in such a way that for known hyperpenalties, such as ridge and lasso, there are software programs that can be used to estimate $\tilde{\lambda}^{(d)}$.

hyperpenalty $\lambda^{(d)}$, $d = 1, \dots, D$ We seek to compute $\hat{\lambda}^{(d)}$ in such a way that the penalized moment estimates $\tilde{\gamma}^{(d)}(\hat{\lambda}^{(d)})$ are stable. Instead of using the computationally difficult and time-consuming approach of cross-validation (CV) on samples, random splits of covariate groups are used. The method of random divisions is as follows: each group $G_g^{(d)}$ is randomly divided into two parts, namely $G_{g,in}^{(d)}$ and $G_{g,out}^{(d)}$. In equation (11), only the inner part of the moment equations is used to solve the linear system (12), with $A_{in}^{(d)}$ and $b_{in}^{(d)}$, depending on which covariates belong to the inner part. Similarly, a linear system is discussed only for the outer part with a matrix and vector corresponding to $A_{out}^{(d)}$ and $b_{out}^{(d)}$. Given that both parts belong to the same group, any stable estimator $\tilde{\gamma}^{(d)}(\lambda^{(d)})$ that adheres to the imposed group structure must correspond to both linear systems corresponding to the inner and outer parts.

Therefore, $\hat{\lambda}^{(d)}$, averaged over several random splits $\mathcal{S}$, is used for the penalized estimate of $\tilde{\gamma}_{in}^{(d)}(\lambda^{(d)})$, from the inner part that best fits the linear equation of the outer part. That is, the estimate of $\hat{\lambda}^{(d)}$ is obtained by minimizing the mean residual sum of squares (RSS):

$$\begin{aligned}
\hat{\lambda}^{(d)} &= \operatorname{argmin}_{\lambda^{(d)}} RSS_{\gamma^{(d)}}(\lambda^{(d)}) \\
&:= \operatorname{argmin}_{\lambda^{(d)}} RSS_{\gamma^{(d)}} \frac{1}{|\mathcal{S}|} \sum_{\mathcal{S}} \|A_{out}^{(d)} \hat{\gamma}_{in}^{(d)}(\lambda^{(d)}) - b_{out}^{(d)}\|_2^2
\end{aligned} \tag{13}$$

3.3.2 Group set weights $d = 1, \dots, D$, $w = (w^1, \dots, w^d)$ with a known $\hat{\gamma}^{(1)}, \dots, \hat{\gamma}^{(d)}$ and $\hat{\tau}_{global}^2$

After estimating the weight of all groups $\hat{\gamma}^{(d)}$, $d = 1, \dots, D$ for each group set separately, we combine the group sets in a linear combination with group set weights of $w = (w^1, \dots, w^D)$. Like the method used for $\hat{\gamma}^{(d)}$, a linear system is created by setting the moment equations like (11) for all G_{total} groups. By replacing the estimates of $\hat{\gamma}^{(d)}$ and rearranging the equations, a linear system of G_{total} groups and solving D of the unknown equation, the weight estimate of the group set $\hat{w}$ is obtained with an ordinary least squares estimate on 0.

$$\begin{aligned}
\tilde{A}^{(d)} w^{(d)} &= b^{(d)} \\
\tilde{A}^{(d)} &:= C^2 \hat{\tau}_{global}^2 \sum_{d=1}^D \hat{\gamma}^{(d)} T_k^{(d)} \\
b^{(d)} &:= E_{Y,\beta}(\hat{\beta}_k^2) - \hat{\nu}_k - [X' \sigma^{-2} X]^2 \\
\hat{w} &= (\tilde{w})_+, \quad \tilde{w} = \underset{w}{\operatorname{argmin}} ||\tilde{A}\tilde{w} - b||_2^2
\end{aligned} \tag{14}$$

Finally, the following algorithm can be used to explain the proposed method in general.

Algorithm 1: semiparametric regression learning for high dimensional data.

1 for $t = 0$

$$\alpha^{(0)}(z_f) = \sum_{i=1}^n l_i(z_f) Y_i \quad \beta^{(0)} = X' X^{-1} X' Y$$

for $t = 0, 1, 2, \dots, T$

$$\beta^{(t+1)EB} = [X' \sigma^{-2} X + \frac{2}{\tau_{k,local}^2}]^{-1} [X' (Y - X \beta^{(t)EB} - \alpha^{(0)}(Z)) + X' \sigma^{-2} X \beta^{(t)EB}]$$

2 starting point: a_{σ^2} , a_τ , a_τ and b_{σ^2} is know

$$\sigma^{2(t+1)MH} \sim IG(\frac{n}{2} + a_{\sigma^2}, \frac{(y - (X \hat{\beta}^{EB} + \alpha^{(t)MH}(Z)))' (y - (X \hat{\beta}^{EB} + \alpha^{(t)MH}(Z)))}{2} + b_{\sigma^2})$$

$$\alpha^{(t+1)MH}(Z) \sim N[\sigma^{-2(t)MH} \{(\tau^{(t)MH} K)^{-1} + \sigma^{-2(t)MH} I\}^{-1} (y - X \hat{\beta}^{(t)EB}), \{(\tau^{(t)MH} K)^{-1} + \sigma^{-2(t)MH} I\}^{-1}]$$

$$\tau^{(t+1)MH} \sim IG(\frac{n}{2} + a_\tau, \{\frac{\alpha^{T(t)MH}(Z) K(\rho)^{-1} \alpha^{(t)MH}(Z)}{2} + b_\tau\}^{-1})$$

3 for $t = 0, 1, \dots, T$

$$\beta^{(t+1)EB} = [X' \hat{\sigma}^{-2(t)MH} X + \frac{2}{\tau_{k,local}^2}]^{-1} [X' (Y - X \beta^{(t)EB} - \hat{\alpha}^{(t)MH}(Z)) + X' \hat{\sigma}^{-2(t)MH} X \beta^{(t)EB}]$$

$$\tau_{global}^{(t+1)EB} = \sqrt{\frac{\pi}{2}} E_{\beta^{(t)EB} | \tau_{global}^2, \hat{\alpha}^{(t)MH}(Z)}(|\beta^{(t)EB}|)$$

for $t = 0, 1, \dots, T$

$$A^{(d)} \gamma^{(d)(t+1)} = b^{(d)}$$

for $t = 0, 1, \dots, T$

$$\tilde{A}^{(d)} w^{(d)(t+1)} = b^{(d)}$$

4 Simulation study

The aim of the simulation is to show the advantage of using hyper-shrinkage on the group weights and group set weights to estimate the unknown parameters when we use the semi-parametric model. The number of samples $n = 100$ and the number of covariates $p = 300$ are assumed in this section. We use three co-data sources for this simulation. The first group is the random non-overlapping group, the second group is informative non-overlapping groups of the same size, and the third group is informative hierarchical groups. Hierarchical and overlapping groups are divided into two groups with all covariates, which in turn are divided into two other groups, and this is repeated for the rest. Hierarchical groups in our proposed method use the combination of hierarchical lasso and ridge hyper-shrinkage to select and estimate the weight

of the prior variances. In this simulation, common random and informative data are defined as follows:

- (1) Random: 300 covariates are randomly assigned to approximately equal groups with $G = 5$.
- (2) Informative: We consider covariates based on the size ranking of each regression coefficient, $k = 1, \dots, p$, $|\beta_k^0|$ where for each pair of the two groups of, $G_i G_j 1 \leq i < j \leq G$, and for $k \in G_i, k \in G_j : |\beta_k^0| < |\beta_{kl}^0|$.

For estimation of parametric component, for each pair of training and test sets, we consider $\sigma^2 = 1$ and $\tau_{k,local}^2 = 0.1$ for the variance parameter in the first step. Considering the prior distributions and the real and simulated regression coefficients, we maximize β according to (3).

$$\begin{aligned} \beta_k^0 &\sim N(0, \tau_{k,local}^2) & k = 1, 2, \dots, p \\ [X_{train}]_{ij}, [X_{test}]_{ij} &\sim N(0, 1), & i = 1, \dots, n, \quad j = 1, \dots, p \\ z_b &\sim N(0, 1) & b = 1, \dots, n \end{aligned}$$

For estimation of non-parametric component, to simulate in this section, we simulated 1000 datasets with two sets of n and p values. In this way, we use $\hat{\beta}$ estimated by the empirical bayes method. To be paid to the parameters (σ^2, α, τ) . In this section, because we use algorithm MH for non-parameter component and variance, fixed values for this algorithm are: $b_{\sigma^2} = 1$ and $a_{\sigma^2} = 2.01$, $b_{\tau} = 10$, $a_{\tau} = 2.0001$ and also the initial values for the start of the algorithm $\sigma^2 = 100$ and $\tau = 0.21$. We consider the two stage introduced for the kernel functions. For each case, we fitted the semi parameter model using our hierarchical Bayesian method. We considered both Gaussian kernel and polynomial kernel with $d = 1, 2$.

First, we examine the results for the case of the Gaussian kernel function and then examine the forms and results of the polynomial function. Figure 1 (a) shows that the informative non-overlapping group set with the same size is the only group set selected in all rows. It shows that this group has more information for prediction than the other two groups. Among the five divided groups, the first group in the random non-overlapping group and the fifth group in the informative non-overlapping group have a group weight higher than 1; thus, they are more important for prediction. Co-data continuous with a lower mean yield a lower prior variance or, equivalently, a larger penalty. In the rest of this section, the performance of estimation and prediction, based on our proposed method, has been compared with Vannee et al. (2021) method. we consider our proposed method under 50 random and informative data and compare it with co-data ridge, linear, gam, scam.p, scam.pmi and AD data.

For more details, refer to Vannee et al. (2021). Figure 2 (a) shows the prior variance estimates for the types of common data models. As expected, the estimated prior variance is constant for random co-data and increases for informative co-data, resulting in a general agnostic ridge having similar prediction performance on random co-data and better prediction performance on informative data. If compared with one another, gam and AD are the same in random mode, but in informative mode, gam has more MSE. Figures

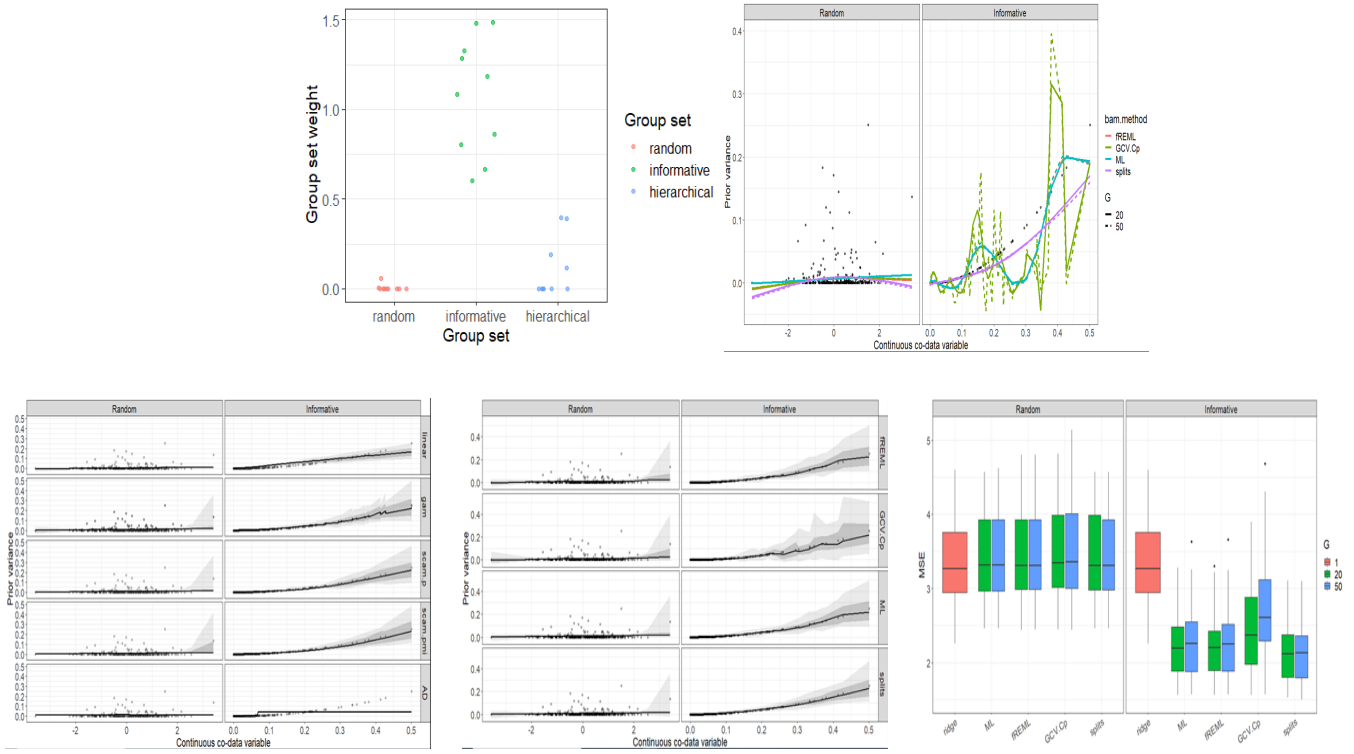


Figure 2: For Gaussian kernel: (a) Estimated prior variance for various co-data models. (b) Estimated prior variance for the generalised additive co-data model using various smoothing parameter estimation methods. (c) boxplots of the MSE of the predictions on the test sets for the ordinary ridge model ($G = 1$ co-data intercept variable) and for a generalised additive co-data model using various smoothing parameter estimation methods and $G = 20$ or 50 splines.

2 (b) and (c) are plotted in a test set with $G = 20$ and $G = 50$. We conclude that except for the generalized cross-validation criterion GCV.Cp, the estimates and corresponding prediction performance appear to be robust (robust) for some splines. This can be seen in figure 1 (b).

For the case of the polynomial kernel function with $r = 1$ and $\rho = 1$, it can be said that the estimated prior variance is constant for random shared data and increases for informative shared data. This results in the conventional agnostic ridge having similar prediction performance on random joint data and better prediction performance on informative data (table 1). If compared, gam and AD are the same in the random, but in the informative, gam has more MSE. We conclude that, except for the generalized cross-validation criterion GCV.Cp, the estimates and the corresponding prediction performance seem to be robust for some splines (Figures 3 (b) and (c)). In table 1, based on the semi-parametric models, all the presented results show a significant reduction in the MSE of the predictions of the simulation data set in the random and informative state for different common data models. Also, the comparison between Gaussian and polynomial functions shows that using the Gaussian kernel function is better for estimating the non-parametric component.

5 An applied study on cervical cancer data

Oncogenic human papillomaviruses (HPVs) are associated with cervical precancerous lesions and the development of cervical cancer. Host genome DNA methylation signatures in normal, precancerous, and cervical tissue may indicate tissue-specific disruption in carcinogenesis. In the article, the authors studied the identification of new candidate genes that are differentially methylated in squamous cell carcinoma compared to DNA samples obtained from grade 3 cervical intraepithelial neoplasia (CIN3) and normal cervical scrapings. Using *IIH450B*[†] method, they detected DNA methylation changes at the genome level in CpG islands, CpG banks and shelves. Their findings show that there is extensive differential methylation in cervical cancer compared to CIN3 or normal cervical tissues. Therefore, in this section, we use the data of this article and using their results as auxiliary data under co-data. For non-parametric data, we used the confidence interval of one of the genes introduced in this research. The co-data includes the following:

1. Extracted P-values where P is the continuous value for each probe from a similar external study. In figure 5 (a), which is about the P values, the groups on the left correspond to the lower P values and are recursively divided into two groups. A hierarchical structure is used on the groups at an additional level of shrinkage to find the discrete one that best fits the data.
2. The CpG Islands: The five non-overlapping groups are based on the genomic annotation of the distance to the nearest CpG island at the CpG site on the DNA. These five groups are arranged in an increasing distance: the CpG Island, North Shore, South Shore, North Shelf, and South Shelf (Figure 5 (b)).

Our aim is to find a classifier that best differentiates normal tissue from CIN3 tissue, a stage with a high risk of progression to cervical cancer in self-collected samples of female cervical tissue. Methylation levels are measured in independent subjects ($n = 64$) with normal tissue (controls) or CIN3 tissue (cases). After filtering, the data contains of methylation levels of $p = 2720$ probes corresponding to specific sites in the DNA. For the hierarchical group, in our proposed method, we use the combination of hierarchical lasso and ridge hypershrinkage to covariate selection and estimate the weight of the previous variances for the hierarchical group. To estimate the penalty function, we also use ridge's estimation as a default. Considering that for the estimation of the non-parametric part, we used the Gaussian kernel function and the polynomial kernel function, at first we analyze the results of each one separately and then compare them with each other.

For Gaussian kernel, figure 4 (a) shows the weight of the group set across folds. P-values group set is the group set that is selected in more sections than CpG-islands group set. It shows that P-values group set has more information for prediction than CpG-Island group set. The Island and S-Shelf have the higher than 1

[†]Illumina Infinium HumanMethylation450 BeadChip

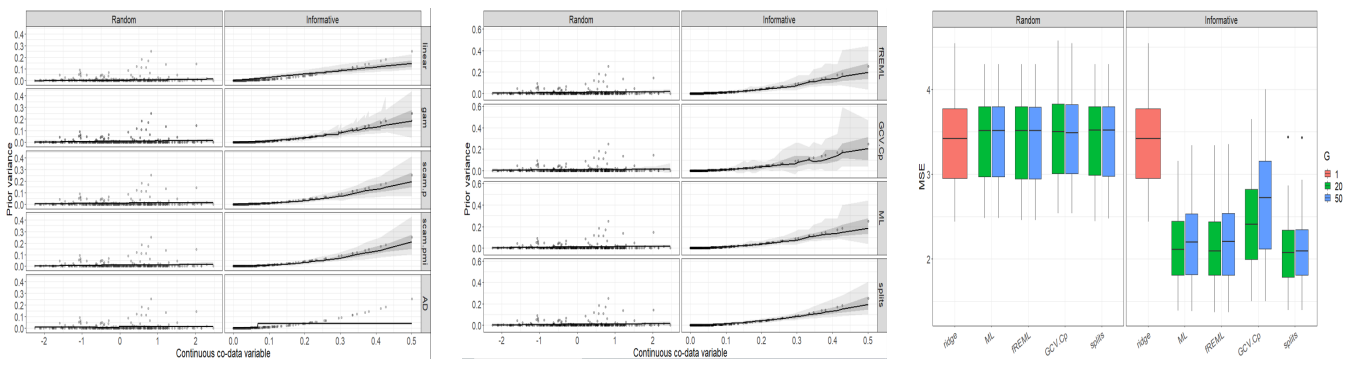


Figure 3: For Polynomial kernel: (a) Estimated prior variance for various co-data models. (b) Estimated prior variance for the generalised additive co-data model using various smoothing parameter estimation methods. (c) boxplots

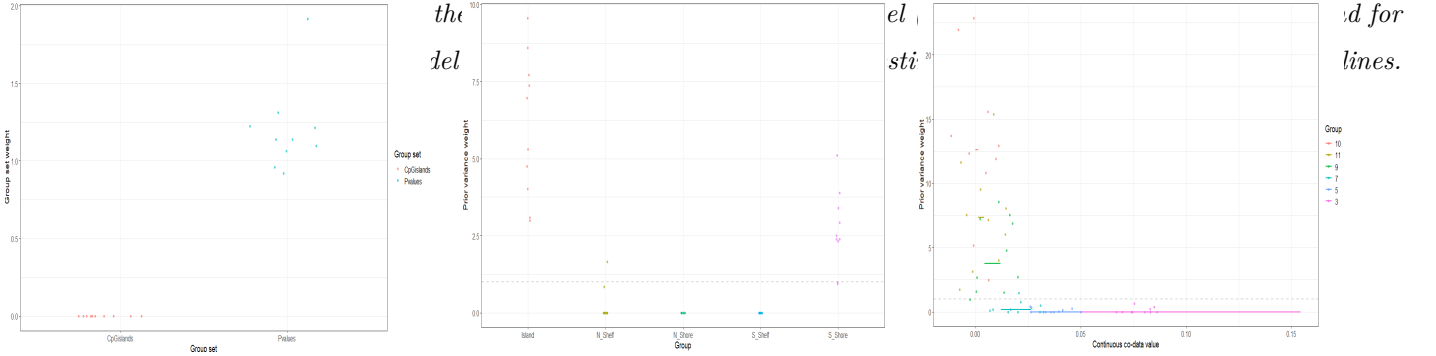


Figure 4: For Gaussian kernel.

prior variance weight and there are more important for prediction, and among these two groups, the island has the highest weight. Groups with a lower average P-value obtain higher prior variance or, equivalently, a lower penalty (Figures 4 (b) and (c)).

For the polynomial kernel, Figure 6 (a) shows that like the Gaussian kernel, P-values group set is group set that is selected in more sections than CpG-islands group set, which indicates that this group set can predict more information than CpG-islands group set. In CpG-islands group set, Island and s-shelf get a group set weight higher than 1 and there are more important for prediction, and among these two groups, the Island has the highest weight. Groups with a lower average P-value obtain a higher prior variance or, equivalently, a lower penalty (Figures 6 (b) and (c)).

In our method, the Gaussian kernel function has the lowest MSE in the estimation of the prior variance of group weights. The comparing this results with Vaneet et al. (2021) shows that the group weight has decreased significantly in all groups. Therefore our proposed method pointed out that, in addition to the advantage of add a level of shrinkage at the group level and use of co-data, add a non-parametric component, use of the Gaussian kernel function increases MSE.

6 Conclusion

We presented a new approach using empirical Bayes and hierarchical Bayes to analyze parameters in semiparametric models. We used multiple and diverse types of co-data to improve prediction and covariate

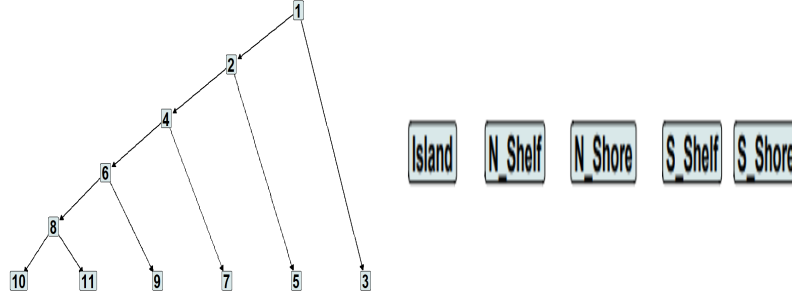


Figure 5

selection for high-dimensional data in penalized semiparametric models. This method allows us to include missing co-data and non-penalized covariates in the calculation. We introduced an additional level of shrinkage at the group level, which provides a unique and flexible framework that accommodates a wide variety of co-data. In addition, this method, deals with the estimation of hyperparameters, but this is done in the presence of the nonparametric component. This opens up new opportunities to investigate the structure of variables at a higher level, allowing us to explore variables in both the parametric and nonparametric components. The advantage of considering the non-parametric component in the results is quite clear, causing a decrease in MSE in the prior variance of group weights compared to the ecpc model, and this issue is closer to the structure of today's data. Based on the non-parametric component, we are allowed to enter other data based on our work context, which will lead to better and more accurate prediction.

Using the ridge penalty for multiple groups and the combination of ridge and lasso hypershrinkage selection, and hierarchical lasso selection for hierarchical co-data in the semiparametric model produced good results. This showed that in both modes of using the Gaussian and polynomial kernel function, the MSE of prior variance weights is significantly reduced compared to the ecpc model. This method can be extended by applying other hypershrinkages to different penalty functions for different co-data types such as graphical data.

We recognize that there are different approaches to analyzing such parameters and that this method may not be theoretically the best method. However, to the best of our knowledge, this is an existing problem, especially in the context of semiparametric models. While the cross-validation was done well in our numerical studies, further research is needed to investigate the optimality of such models. In terms of suggestions for further research, it is suggested that different penalty functions can be used in the penalty model (3), and it is even recommended that the penalty function be based on a non-parametric parameter or a function of both.

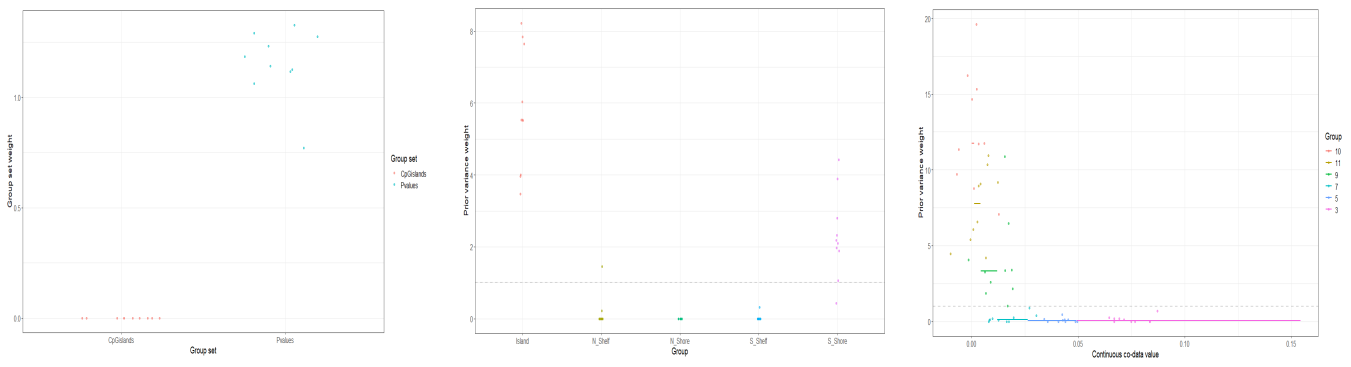


Figure 6: *Polynomial kernel.*

Table 1: *comparison minimum and maximum of boxplots of the the MSE of the predictions on the test sets for various co-data models in Semiparametric model in mode Gaussian kernel and Polynomial kernel with ecpc model.*

λ	Gaussian kernel		Polynomial kernel		ecpc	
	Random	Informative	Random	Informative	Random	Informative
ridge	2.97-3.75	2.99-3.75	2.99-3.75	2.99-3.74	15.59-22.56	17.59-22.56
linear	2.98-3.80	1.75-2.04	2.99-3.74	1.75-2.25	17.6-22.5	11-13.5
gam	2.98-3.90	1.80-2.03	3-3.77	1.80-2.30	17.6-22.5	11-14
scam.p	2.99-3.85	1.80-2.04	3-3.74	1.85-2.40	17.6-22.5	11.75-15
scam.pm	2.98-3.75	1.80-2.04	3-3.74	1.80-2.40	17.6-22.5	11.75-15
AD	3.01-3.80	2.01-2.56	3-3.75	2.13-2.75	17.59-22.5	12-15

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